

When Context Is a Lie About Itself: Pseudo-Context Detection and Neuro-Symbolic Routing for Bengali LLM Hallucination Detection

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Abstract—We describe the system we submitted to the **অলীকবচন** Bengali LLM Hallucination Detection Challenge, a binary faithful-vs-hallucinated classification task scored by F1 on the hallucinated class. Our central finding is diagnostic rather than architectural: roughly 430 of 2,516 test rows ($\approx 17\%$) carry a *pseudo-context* — a content-free meta-sentence that restates the question and asserts an answer exists without providing it. On these rows the gold label is closed-book factual correctness, not context-groundedness, and any reference-grounded verifier systematically inverts its predictions. We therefore treat the benchmark as two tasks hidden under one label and route each row to the cheapest sufficient tool: deterministic Bengali grammar solvers and gold-key joins where the answer is derivable, and a single Platt-calibrated logit-margin pass of Qwen2.5-14B-Instruct where it is not. A corruption-aware repair stage additionally recovers a systematic comma-split truncation artifact in the test questions. Every layer was admitted only after passing a pre-committed gate on the official 299-row labeled sample; layers that failed were discarded and are reported here. The final system reaches 0.937 F1 on the hallucinated class, runs offline in 3.15 h on dual T4 GPUs, and reproduced its submitted predictions zero-diff under the organizers’ rerun protocol.

Index Terms—hallucination detection, Bengali NLP, low-resource languages, neuro-symbolic systems, LLM evaluation

I. INTRODUCTION

Hallucination — output that contradicts provided context or established fact — is a first-order obstacle to deploying large language models (LLMs), and detection resources for Bengali, the world’s sixth most spoken language, have appeared only very recently [1], [5]. The **অলীকবচন** challenge operationalizes detection as binary classification over triples (*context, prompt, response*): label 1 if the response is faithful, 0 if hallucinated, scored by F1 on the hallucinated class ($F1_0$).

The natural reading of this schema is reference-grounded verification: check the response against the context. Our system began there, with a calibrated LLM judge. Its errors, however, were not uniformly distributed. On several linguistically defined families — **সমাস** (compound analysis), **সন্ধি** (euphonic combination), antonyms, idioms, prefixes — the judge’s faithful-rate collapsed far below the $\sim 50\%$ rate

a contrastive benchmark generator produces by construction (Section III). Inspection showed the context on these rows never contains the answer: it is a meta-sentence *about* the existence of an answer. A grounded verifier, behaving correctly by its own logic, votes “unsupported” and is wrong half the time. We call this *pseudo-context*, and we argue it is a task-definition property of the slice, not a model-quality defect — a distinction with direct consequences for how such benchmarks should be attacked and built.

Our contributions are: (i) identification and quantification of the pseudo-context phenomenon, and a routing principle — grounded verification and closed-book correctness require different machinery on different rows; (ii) a neuro-symbolic pipeline in which nine deterministic stages surround a single 14B forward pass, including token-based Bengali morphological solvers (we show that regex word boundaries fail on Bengali combining marks) and a corruption-aware input-repair stage that recovers a systematic comma-split truncation artifact in the test questions; (iii) a gated experimental methodology — every layer validated on the official 299-row sample against pre-committed kill criteria — together with an honest account of the approaches that failed.

II. RELATED WORK

BenHalluEval [1] established the first dedicated hallucination benchmark for Bengali, using a dual-track protocol that separately penalizes false alarms on ground-truth answers and misses on hallucinated candidates; the present challenge is in its design lineage. The organizers likewise draw on a large expert-annotated corpus of Bengali idioms [2], whose evaluation shows no current LLM exceeding 50% accuracy on figurative meaning — consistent with our decision to handle **বাগধারা** rows symbolically rather than by judge. BLUCK [3] demonstrates that LLMs remain weakest on exam-style Bengali linguistics categories such as phonetics — the register from which the pseudo-context families here are drawn. Work on Bengali entailment [4] cautions that Bangla-native encoders do not automatically win on inference-shaped subtasks, sup-

TABLE I

JUDGE-ONLY FAITHFUL-RATE BY PSEUDO-CONTEXT FAMILY, AGAINST THE $\sim 50\%$ RATE IMPLIED BY CONTRASTIVE BENCHMARK CONSTRUCTION. THE GROUNDED JUDGE SYSTEMATICALLY UNDER-LABELS “FAITHFUL” WHEREVER THE CONTEXT CANNOT ENTAIL THE ANSWER; সন্ধি IS THE EXCEPTION BECAUSE THE SURFACE FORM ITSELF CARRIES THE CUE.

Family	n	Judge faithful (%)
সমাস (compound analysis)	49	12.2
বিপরীত শব্দ (antonyms)	52	30.8
বাগধারা (idioms)	30	26.7
সন্ধি (euphonic combination)	42	81.0

porting our choice to settle backbone questions empirically rather than by prior. General hallucination benchmarks such as HaluEval [5] frame detection as grounded verification; our result shows that mixed benchmarks can contain per-row exceptions to that frame, and that detecting the frame is itself part of the task.

III. THE PSEUDO-CONTEXT PHENOMENON

Approximately 430 test rows ($\approx 17\%$) share a template: the prompt asks a closed-book Bengali-grammar or vocabulary question, and the context is a meta-sentence such as “বাংলা ব্যাকরণ অনুযায়ী শব্দটির একটি সুনির্দিষ্ট বিপরীত শব্দ রয়েছে” (“according to Bengali grammar, the word has a definite antonym”) — asserting an answer exists while providing no evidence for or against the response. For সমাস rows the context does contain usable signal (the ব্যাসবাক্য, the expansion phrase), but the answer must be *derived* from it by grammatical rule, not retrieved.

Table I quantifies the inversion. A contrastive generator does not construct 49 সমাস items and make 84% of them wrong; when we derive the answer from each ব্যাসবাক্য with a deterministic classifier (Section IV-A), 25 of 49 responses (51%) are grammatically correct — precisely the balanced split the generator intends. The judge’s 12.2% is therefore not noise but a signature: on these rows it is answering a different question (“does the context entail the response?”) than the benchmark is asking (“is the response correct?”). Prompt-side fixes cannot repair this, because the correct behavior is family-dependent; the repair must be architectural. We note this failure mode is invisible to aggregate metrics until the family structure is recovered, which we did by clustering context templates — a step we recommend as standard practice for mixed benchmarks.

IV. METHOD

Figure 1 shows the pipeline. The design principle throughout is *cheapest sufficient tool with calibrated abstention*: every deterministic stage returns a verdict only inside its verified competence and otherwise abstains, deferring to the judge; the single neural pass is reserved for rows no symbolic mechanism can decide. The execution order is load-bearing and is preserved exactly in the submitted notebook.

A. Deterministic Rule Layer

The first stage decides rows where the label is a deterministic function of the raw inputs joined against declared external resources. Gold-key joins normalize prompt and answer text (NFC normalization, zero-width character removal, digit-script unification) and match exact keys against BenHalluEval-derived QA pools, TyDiQA-Bengali [9], and bangla-mmlu; a response agreeing with the retrieved gold answer is labeled faithful, a response matching a known distractor is labeled hallucinated, and ambiguous families (WEAK_A) abstain by construction. An idiom arm resolves বাগধারা rows against a figurative-meaning pool.

The grammar solver handles the two families where the answer is derivable. For সমাস, an 18-rule token-based classifier maps the quoted ব্যাসবাক্য to its compound type (e.g., দ্বন্দ্ব, দ্বিগু, বহুব্রীহি) and compares against the response. For সন্ধি, the combined form always differs from naive concatenation of the two quoted words unless the context states অপরিবর্তিত (unchanged), so equality with the concatenation implies hallucination. One implementation finding is worth recording for low-resource practitioners: regex \b word boundaries silently fail on Bengali combining marks, so all matching is token-based. The solver fires on exactly 91 rows (49 + 42), is base-independent, and contributes 44 label flips.

B. Calibrated Logit-Margin Judge

Rows on which every rule abstains are decided by Qwen2.5-14B-Instruct [6]. Rather than sampling generations, we read the logit margin between the faithful and hallucinated verdict tokens in a single forward pass, making the judge deterministic and ~ 2 s/row. Margins are mapped to probabilities by Platt scaling [7] fit *live* on the official 299-row labeled sample inside the notebook, with decision threshold $\tau = 0.340$ selected on the same sample under the $F1_0$ objective; because the metric weighs a recovered miss $\approx 1.23 \times$ a corrected false alarm, the threshold sits deliberately on the sensitive side. The judge alone reaches $F1_0 = 0.762$ on the anchor sample.

C. Corpus Consensus Arms and Corrective Reverts

Three arms join judge-decided rows against a declared pattern corpus using unanimity-filtered keys: exact-pair faithful (S1, 98 fires), exact-pair hallucinated restricted to no-context rows (S2, 38 fires), and a question-level faithful join (S3, 85 fires) guarded against grammar-family prompts. Because corpus precision (≈ 0.65) is below the judge’s accuracy in its confident bands (0.846), a corrective revert stage then undoes exactly the low-precision corpus flips: rows flipped 0 $\rightarrow$ 1 by S1/S3 whose response fails a containment check against a present context (ctx_guard, 23 rows), and S3 flips contradicted by strong hallucination evidence (R1b, 34 rows; overlap 8; 49 rows total reverted), with a historical dead-rule exemption ($\sim$ R1f) preserved for exact reproducibility. Pseudo-context families and solver decisions are protected from reversal — the routing principle applied a second time.

□ deterministic □ neural (one pass) □ corrective

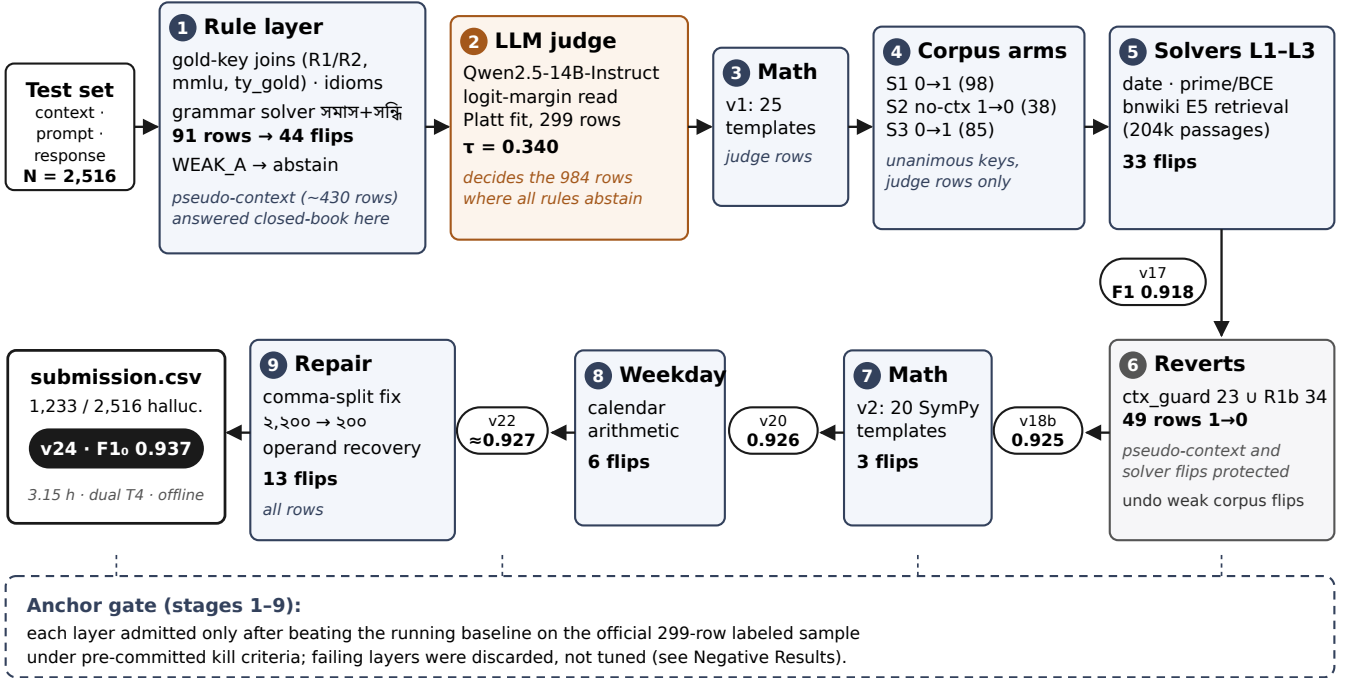


Fig. 1. The v24 inference pipeline. Stages execute in numbered order (the ordering is load-bearing: reverts must precede math v2, weekday, and repair). Blue-gray denotes deterministic symbolic components; the warm box is the single neural pass; the dashed band is the anchor-gate methodology applied to every stage.

D. Specialized Verifiers

Where the response is checkable by computation, we compute. Two template banks parse Bengali arithmetic word problems (25 regex templates; then 20 SymPy-verified templates) and override the judge with the exact answer; a weekday solver performs calendar arithmetic; small verifiers check dates, primality, and BCE conventions; and a retrieval layer scores factoid responses against 204,018 Bengali Wikipedia passages embedded with multilingual-e5-base [8] (embeddings precomputed and attached, so inference stays offline). Together these contribute 33 + 3 + 6 flips at distinct pipeline positions.

E. Corruption-Aware Input Repair

Auditing solver failures exposed a systematic artifact in test questions 183–241: numerals containing Bengali comma separators are truncated at the comma (২,২০০ → ২০০), corrupting work-rate, price, and interest word problems. The final stage detects affected templates and recovers the intended operand by bisection over candidate restorations, accepting a repair only when exactly one candidate makes the problem internally consistent, then re-solves. This stage contributes 13 flips and, we emphasize, is a deterministic function of the raw test text — no row identifiers or manual labels are involved, as required by the competition’s reproducibility rules.

TABLE II
LEADERBOARD F1₀ PROGRESSION. EACH ROW NAMES THE STAGE ADDED;
ORDERING MATCHES FIG. 1.

Ver.	Added component	F1 ₀
v17	rules + judge + math v1 + corpus + solvers	0.918
v18b	corrective reverts (49 rows)	0.925
v20	math v2 (SymPy)	0.926
v22	weekday arithmetic	≈0.927
v24	corruption-aware repair	0.937

V. EXPERIMENTS AND RESULTS

A. Gated Development Protocol

All development was governed by a single rule: a layer enters the pipeline only if it improves F1₀ on the official 299-row labeled sample over the running baseline, with kill criteria fixed *before* building. Leaderboard submissions (4/day) were spent confirming anchor-validated changes, never searching; no test row was ever manually labeled.

B. Results

Table II traces the score. Two observations. First, the largest single step after the base system is the input-repair stage (+0.009): on a benchmark about detecting corrupted *outputs*, the decisive residual errors were corrupted *inputs* — a reminder that data forensics belongs in the error-analysis loop. Second, gains came overwhelmingly from deciding *which*

mechanism answers a row, not from strengthening the neural component: a 32B judge variant gained only +0.003 and was rejected (Section VI).

C. Reproducibility

The submitted notebook rebuilds the final predictions from the raw test file end-to-end: the judge is re-run, Platt calibration is re-fit on the anchor sample, and every downstream stage is a pure function of raw text plus declared external files. A clean top-to-bottom execution with internet disabled reproduced the submitted CSV with zero differing rows in 3.15h on dual T4 GPUs (projected ≈ 3.4 h on the 5,000-row held-out fold), well inside the competition budget; the organizers’ own two-run protocol subsequently confirmed reproduction. Determinism comes cheaply here: the only stochastic component in most pipelines — sampled generation — is absent by design.

VI. NEGATIVE RESULTS

We report what failed, since the gates that killed these are as much a part of the method as the stages that shipped. A 32B judge gained +0.003 but its generating code could not be packaged reproducibly — rejected. A prompt ensemble cost −0.007. Few-shot *generative* judging with an AWQ-quantized 14B scored F_1 0.717 on the anchor, versus 0.814 for the margin-based configuration then in use, and overran the time budget (13.2h) — sampling added noise, not signal. A disagreement veto deferring gold-key rules to saturated judge margins collapsed the anchor from 0.888 to 0.828 — killed by its pre-committed criterion despite addressing a real defect. QLoRA fine-tuning failed on three environment configurations. Extending pseudo-context routing beyond the solver-covered families was refuted by anchor evidence: the phenomenon is real, but exploiting it requires a mechanism that can actually derive the closed-book answer.

VII. LIMITATIONS

(i) A small number of externally sourced gold-key entries are themselves wrong and force incorrect labels on specific rows; our attempted fix failed its anchor gate, so the defect ships documented rather than patched. (ii) The judge’s discrimination is weakest exactly where no context exists (AUC 0.748 on no-context rows), bounding closed-book performance outside solver coverage. (iii) One test item’s ground truth is linguistically contested (মধ্যপদলোপী vs. উপমান কর্মধারয় for রক্তকমল); we reported it publicly on the competition forum per the rules. (iv) The symbolic layers are Bengali-specific by design; the routing principle transfers, the solvers do not. (v) All development-time gating relied on a single 299-row labeled sample; gates this small can admit or kill a layer on noise, which is why we paired them with pre-committed criteria rather than repeated peeking.

VIII. CONCLUSION

The system’s score rests on a diagnosis: this benchmark interleaves grounded verification with closed-book correctness at the row level, and the two demand different machinery.

Routing rows to deterministic Bengali grammar solvers, computational verifiers, and evidence joins — reserving a single calibrated 14B forward pass for the genuinely open rows — achieved 0.937 F_1 , fully offline and exactly reproducible. We suggest two takeaways for low-resource evaluation generally: mixed benchmarks should label (or at least document) which task frame each row belongs to, and detection systems should treat identifying the frame as part of the task rather than an exception to it.

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